

Quality Report



Generated with Pix4Dmapper version 4.6.4



Important: Click on the different icons for:



Help to analyze the results in the Quality Report



Additional information about the sections



Click [here](#) for additional tips to analyze the Quality Report

Summary



Project	ejercicio
Processed	2026-09-03 09:57:38
Camera Model Name(s)	FC6310_8.8_4864x3648 (RGB)
Average Ground Sampling Distance (GSD)	2.50 cm / 0.99 in
Area Covered	0.378 km ² / 37.8280 ha / 0.15 sq. mi. / 93.5233 acres
Time for Initial Processing (without report)	10m:20s

Quality Check



Images	median of 66432 keypoints per image	
Dataset	274 out of 275 images calibrated (99%), all images enabled	
Camera Optimization	11.4% relative difference between initial and optimized internal camera parameters	
Matching	median of 18925.6 matches per calibrated image	
Georeferencing	yes, no 3D GCP	

Preview

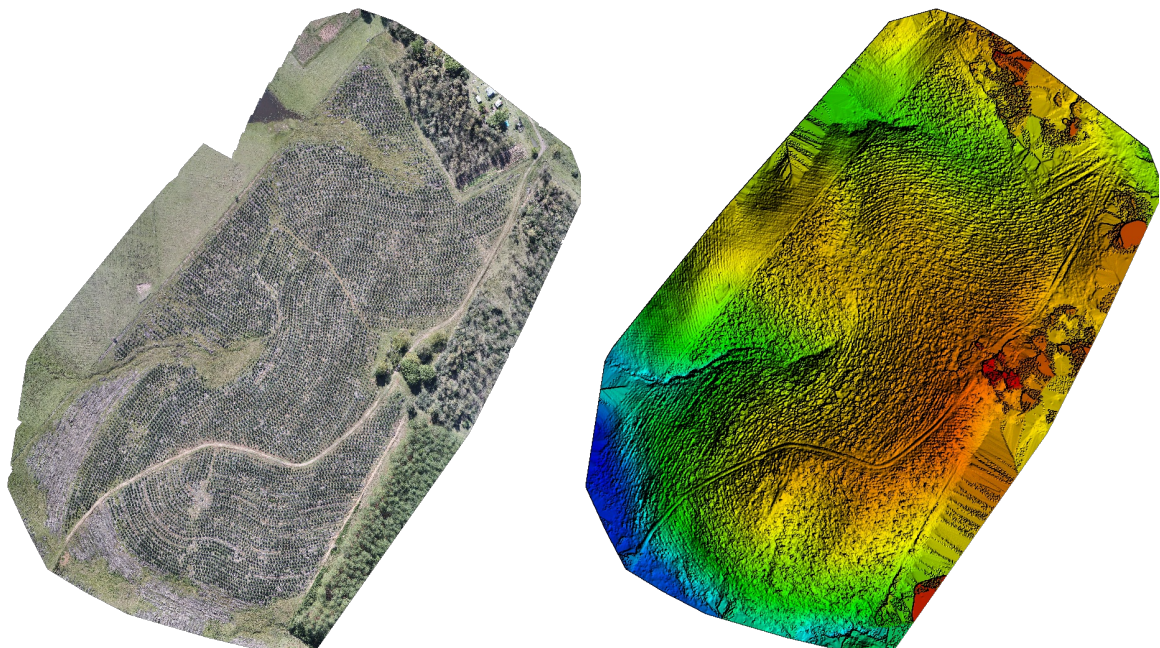


Figure 1: Orthomosaic and the corresponding sparse Digital Surface Model (DSM) before densification.

Calibration Details



Number of Calibrated Images	274 out of 275
Number of Geolocated Images	275 out of 275

Initial Image Positions

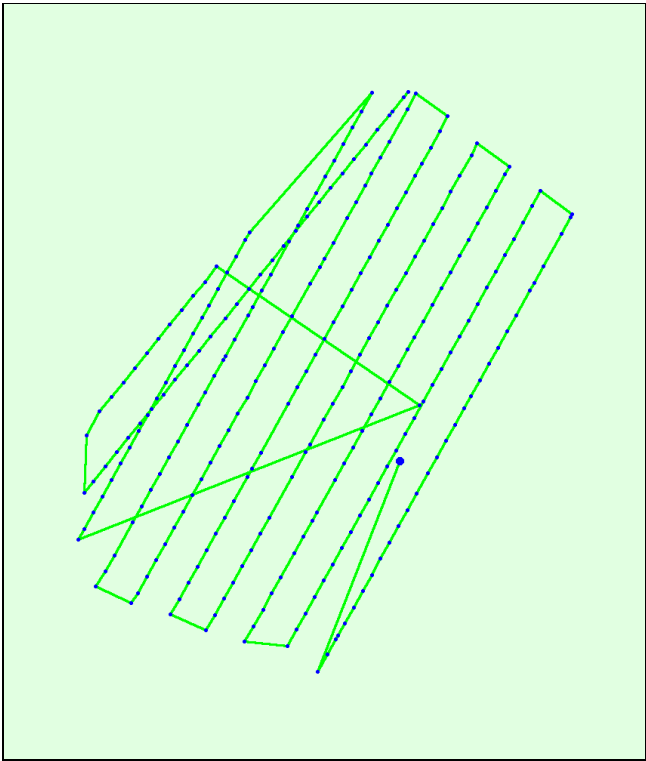
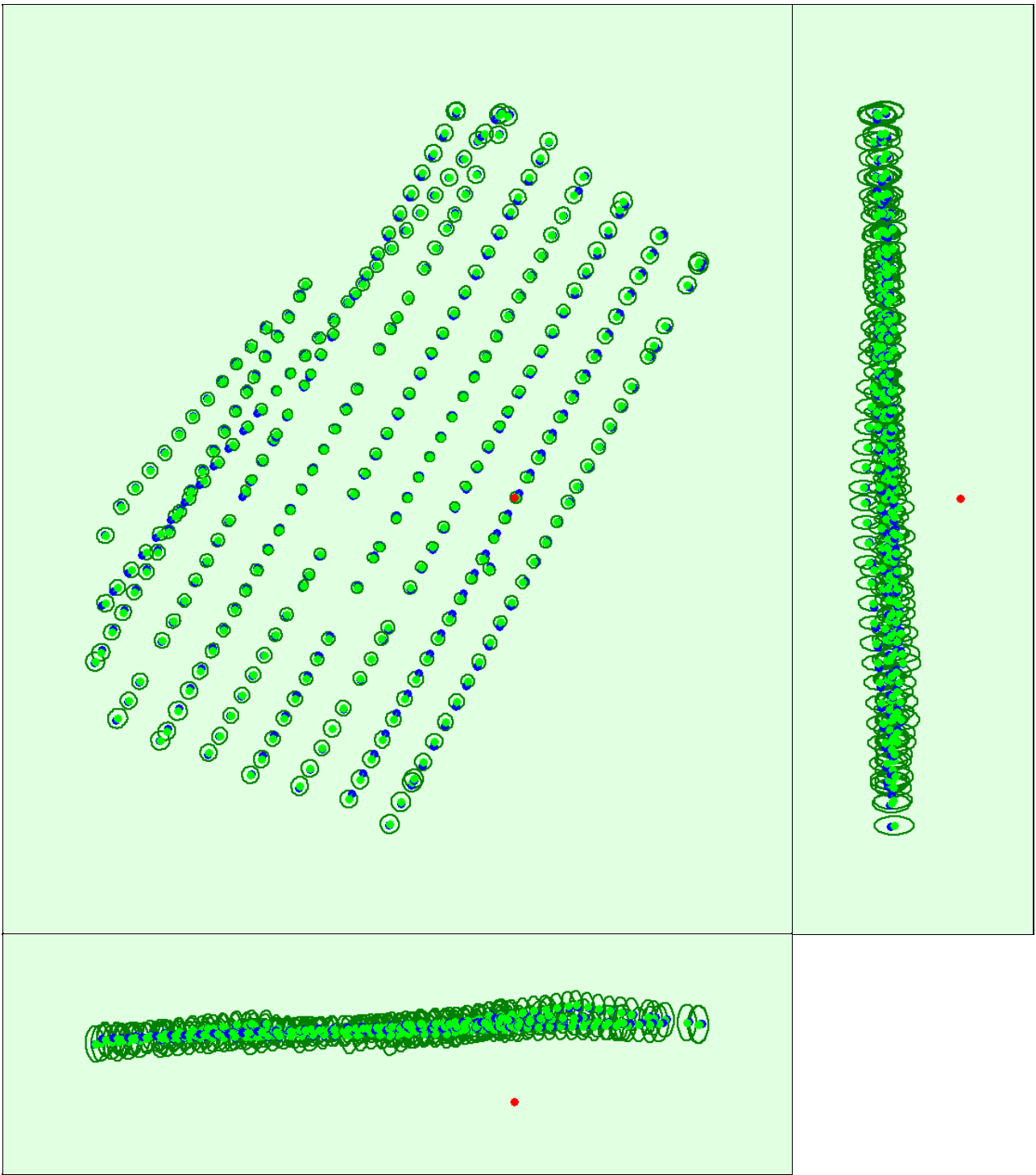


Figure 2: Top view of the initial image position. The green line follows the position of the images in time starting from the large blue dot.

Computed Image/GCPs/Manual Tie Points Positions





Uncertainty ellipses 50x magnified

Figure 3: Offset between initial (blue dots) and computed (green dots) image positions as well as the offset between the GCPs initial positions (blue crosses) and their computed positions (green crosses) in the top-view (XY plane), front-view (XZ plane), and side-view (YZ plane). Red dots indicate disabled or uncalibrated images. Dark green ellipses indicate the absolute position uncertainty of the bundle block adjustment result.

? Absolute camera position and orientation uncertainties



	X[m]	Y[m]	Z[m]	Omega [degree]	Phi [degree]	Kappa [degree]
Mean	0.149	0.150	0.297	0.055	0.059	0.025
Sigma	0.025	0.026	0.045	0.007	0.004	0.000

? Overlap



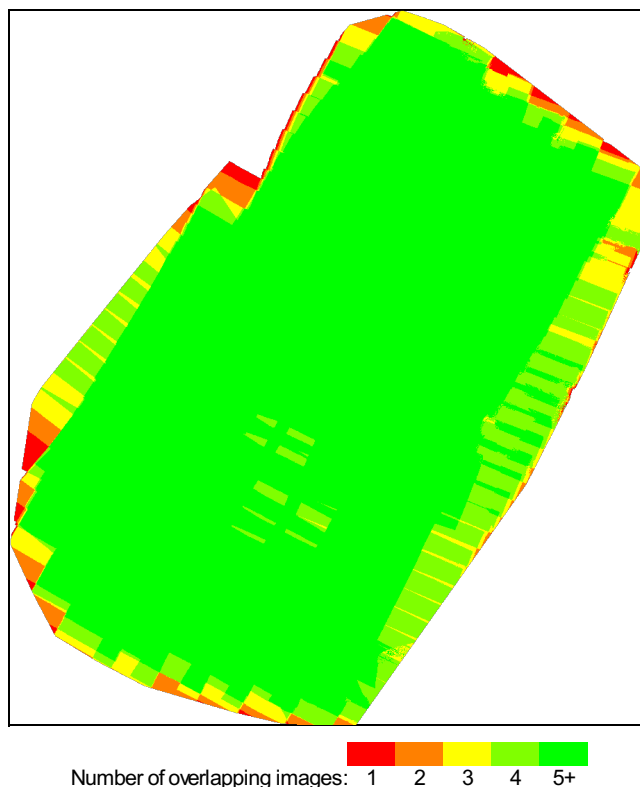


Figure 4: Number of overlapping images computed for each pixel of the orthomosaic. Red and yellow areas indicate low overlap for which poor results may be generated. Green areas indicate an overlap of over 5 images for every pixel. Good quality results will be generated as long as the number of keypoint matches is also sufficient for these areas (see Figure 5 for keypoint matches).

Bundle Block Adjustment Details



Number of 2D Keypoint Observations for Bundle Block Adjustment	5240727
Number of 3D Points for Bundle Block Adjustment	2120225
Mean Reprojection Error [pixels]	0.167

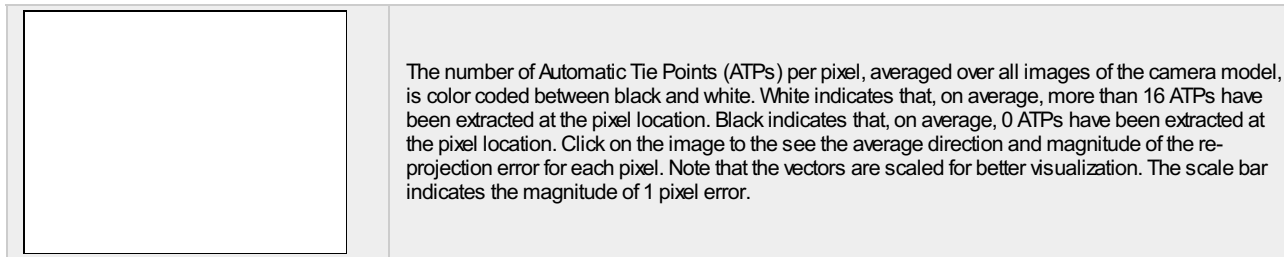
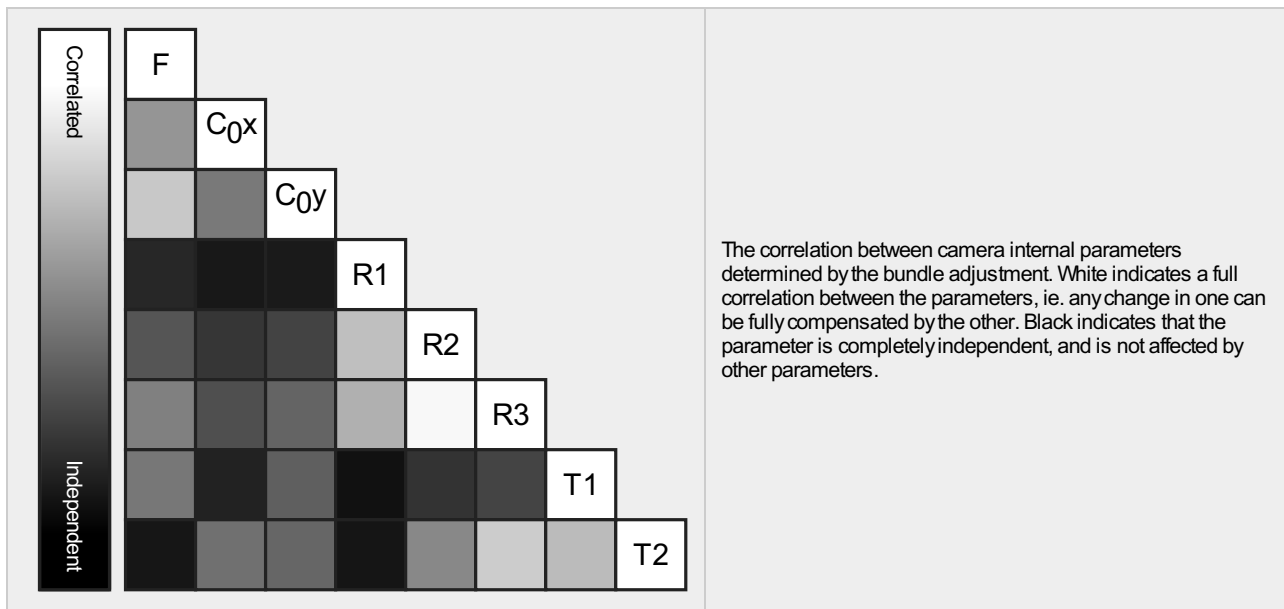
? Internal Camera Parameters

FC6310_8.8_4864x3648 (RGB). Sensor Dimensions: 11.407 [mm] x 8.556 [mm]



EXIF ID: FC6310_8.8_4864x3648

	Focal Length	Principal Point x	Principal Point y	R1	R2	R3	T1	T2
Initial Values	3666.955 [pixel] 8.600 [mm]	2432.001 [pixel] 5.704 [mm]	1823.999 [pixel] 4.278 [mm]	0.004	-0.017	0.019	-0.000	0.000
Optimized Values	4085.250 [pixel] 9.581 [mm]	2438.952 [pixel] 5.720 [mm]	1844.320 [pixel] 4.325 [mm]	0.005	-0.030	0.040	-0.000	0.000
Uncertainties (Sigma)	7.909 [pixel] 0.019 [mm]	0.200 [pixel] 0.000 [mm]	0.232 [pixel] 0.001 [mm]	0.000	0.001	0.001	0.000	0.000



2D Keypoints Table



	Number of 2D Keypoints per Image	Number of Matched 2D Keypoints per Image
Median	66432	18926
Mn	46605	3113
Max	79856	36725
Mean	65801	19127

3D Points from 2D Keypoint Matches



	Number of 3D Points Observed
In 2 Images	1576268
In 3 Images	327051
In 4 Images	109533
In 5 Images	46564
In 6 Images	26809
In 7 Images	15828
In 8 Images	8822
In 9 Images	4083
In 10 Images	2588
In 11 Images	1460
In 12 Images	880
In 13 Images	205
In 14 Images	77
In 15 Images	43
In 16 Images	13
In 17 Images	1

2D Keypoint Matches



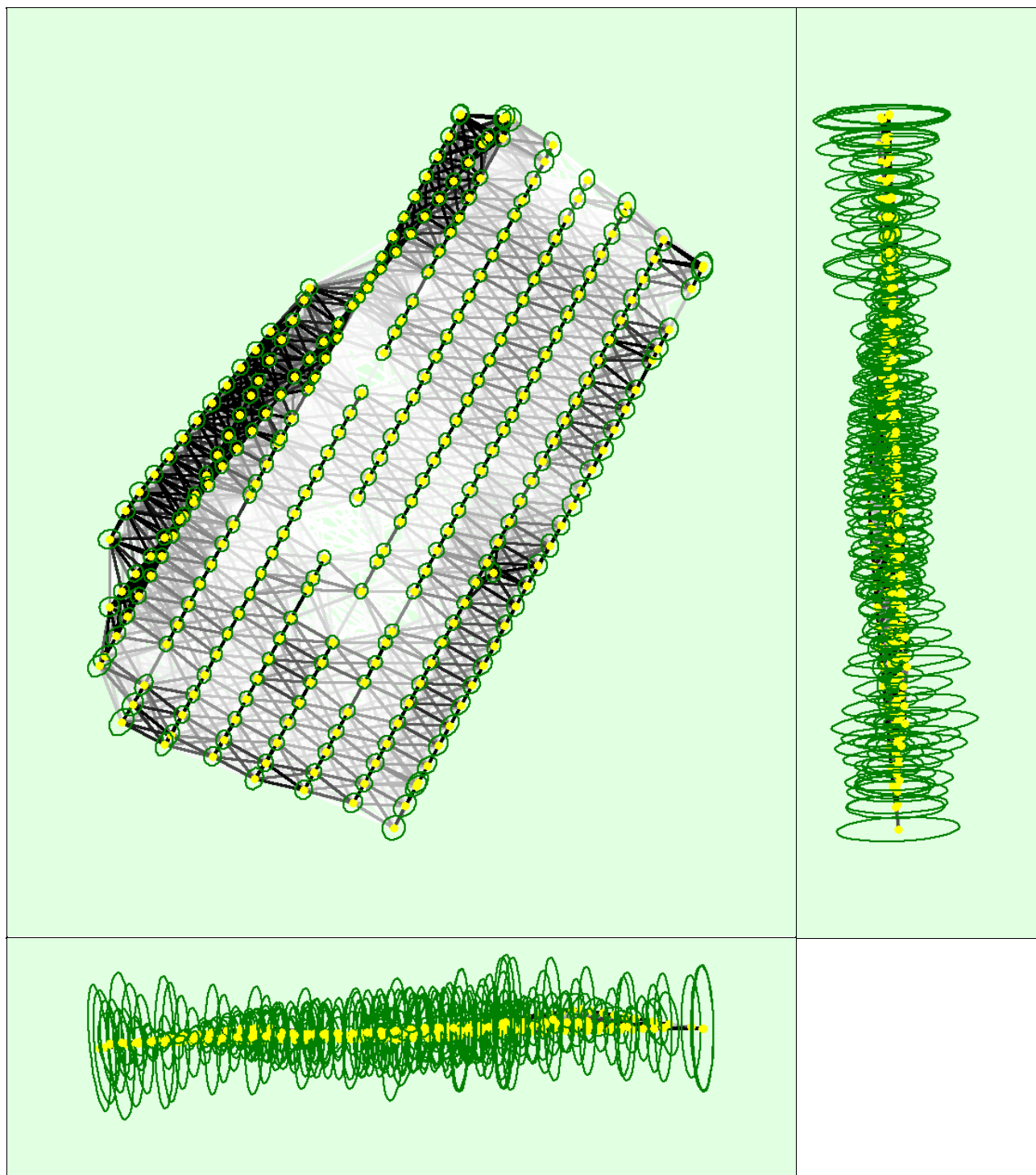


Figure 5: Computed image positions with links between matched images. The darkness of the links indicates the number of matched 2D keypoints between the images. Bright links indicate weak links and require manual tie points or more images. Dark green ellipses indicate the relative camera position uncertainty of the bundle block adjustment result.

Relative camera position and orientation uncertainties



	X[m]	Y[m]	Z[m]	Omega [degree]	Phi [degree]	Kappa [degree]
Mean	0.016	0.018	0.056	0.033	0.022	0.003
Sigma	0.003	0.003	0.033	0.017	0.012	0.001

Geolocation Details



Absolute Geolocation Variance



Min Error [m]	Max Error [m]	Geolocation Error X [%]	Geolocation Error Y [%]	Geolocation Error Z [%]
-	-15.00	0.00	0.00	0.00
-15.00	-12.00	0.00	0.00	0.00
-12.00	-9.00	0.00	0.00	0.00
-9.00	-6.00	0.00	0.00	0.00
-6.00	-3.00	5.47	17.88	0.00
-3.00	0.00	39.78	33.58	54.38
0.00	3.00	47.81	35.77	45.26
3.00	6.00	6.93	9.49	0.36
6.00	9.00	0.00	3.28	0.00
9.00	12.00	0.00	0.00	0.00
12.00	15.00	0.00	0.00	0.00
15.00	-	0.00	0.00	0.00
Mean [m]		0.000253	0.000126	0.000016
Sigma [m]		1.781945	2.751821	0.971751
RMS Error [m]		1.781945	2.751821	0.971751

Min Error and Max Error represent geolocation error intervals between -1.5 and 1.5 times the maximum accuracy of all the images. Columns X, Y, Z show the percentage of images with geolocation errors within the predefined error intervals. The geolocation error is the difference between the initial and computed image positions. Note that the image geolocation errors do not correspond to the accuracy of the observed 3D points.

Relative Geolocation Variance



Relative Geolocation Error	Images X [%]	Images Y [%]	Images Z [%]
[-1.00, 1.00]	100.00	93.43	100.00
[-2.00, 2.00]	100.00	100.00	100.00
[-3.00, 3.00]	100.00	100.00	100.00
Mean of Geolocation Accuracy [m]	5.000000	5.000000	10.000000
Sigma of Geolocation Accuracy [m]	0.000000	0.000000	0.000000

Images X, Y, Z represent the percentage of images with a relative geolocation error in X, Y, Z.

Geolocation Orientational Variance	RMS [degree]
Omega	0.780
Phi	0.680
Kappa	4.574

Geolocation RMS error of the orientation angles given by the difference between the initial and computed image orientation angles.

Initial Processing Details



System Information



Hardware	CPU: Intel(R) Core(TM) i7-9700K CPU @ 3.60GHz RAM: 32GB GPU: NVIDIA GeForce RTX 2070 SUPER (Driver: 32.0.16.1656)
Operating System	Windows 10 Pro, 64-bit

Coordinate Systems



Image Coordinate System	WGS 84 (EGM96 Geoid)
Output Coordinate System	WGS 84 / UTMzone 21S (EGM96 Geoid)

Processing Options



Detected Template	3D Maps
Keypoints Image Scale	Full, Image Scale: 1
Advanced: Matching Image Pairs	Aerial Grid or Corridor
Advanced: Matching Strategy	Use Geometrically Verified Matching: no
Advanced: Keypoint Extraction	Targeted Number of Keypoints: Automatic
Advanced: Calibration	Calibration Method: Standard Internal Parameters Optimization: All External Parameters Optimization: All Rematch: Auto, yes

Point Cloud Densification details



Processing Options



Image Scale	multiscale, 1/2 (Half image size, Default)
Point Density	Optimal
Minimum Number of Matches	3
3D Textured Mesh Generation	yes
3D Textured Mesh Settings:	Resolution: Medium Resolution (default) Color Balancing: no
LOD	Generated: no
Advanced: 3D Textured Mesh Settings	Sample Density Divider: 1
Advanced: Image Groups	group1
Advanced: Use Processing Area	yes
Advanced: Use Annotations	yes
Time for Point Cloud Densification	25m:31s
Time for Point Cloud Classification	NA
Time for 3D Textured Mesh Generation	12m:19s

Results



Number of Generated Tiles	4
Number of 3D Densified Points	40275630
Average Density (per m ³)	155.38

DSM, Orthomosaic and Index Details



Processing Options



DSM and Orthomosaic Resolution	1 x GSD (2.5 [cm/pixel])
DSM Filters	Noise Filtering: yes Surface Smoothing: yes, Type: Sharp
Raster DSM	Generated: yes Method: Inverse Distance Weighting Merge Tiles: yes
Orthomosaic	Generated: yes Merge Tiles: yes GeoTIFF Without Transparency: no Google Maps Tiles and KML: no
Time for DSM Generation	39m:59s
Time for Orthomosaic Generation	40m:19s
Time for DTM Generation	00s

Time for Contour Lines Generation	00s
Time for Reflectance Map Generation	00s
Time for Index Map Generation	00s